

Heat transport in chaotic magnetic fields

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1 PyOculus workflow

In PyOculus, we can define a magnetic configuration by specifying the components of the magnetic field $\mathbf{B}$. In our case, we want to study the following magnetic field in slab geometry, in which a perturbation creates an island structure:

$$\begin{cases} B_x = m\epsilon(x) \sin \alpha \\ B_y = \iota'(x - x_0) + \epsilon'(x) \cos \alpha \\ B_z = 1 \end{cases} \quad (1)$$

For a system with $x \in [0, 1]$ containing an island at $x_s = 0.5$, as a simple model for a perturbation localized around the rational surface, we choose

$$\epsilon(x) = \delta \cdot x(1 - x). \quad (2)$$